ONLINE LEAVE MANAGEMENT SYSTEM

Minor Project – II Report

Submitted

In partial fulfillment of the requirements for the award of the diploma

DIPLOMA

IN

COMPUTER SCIENCE & ENGINEERING

By

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Under the Guidance of

Mr. A. Siva Rao

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Directorate of Polytechnic Education

VIGNAN'S FOUNDATION FOR SCIENCE, TECHNOLOGY&RESEARCH

(Deemed to be University)

Vadlamudi, Guntur -522213

DIRECTORATE OF POLYTECHNIC EDUCATION

CERTIFICATE

This is to certify that the Minor Project – II Report entitled **"Online Leave Management System"** is submitted by **"SK. Karishma (231FE04042)** in the partial fulfilment of Minor Project - II work carried out in the Directorate of Polytechnic Education, VFSTR Deemed to be University.

Mr. A. Suresh Babu	**Dr. S.K.Satpathy**
Assistant Professor	Director
	VFSTR

External Examiner

DECLARATION

I hereby declare that the project entitled **"Online Leave Management System"** submitted for the the Directorate of Polytechnic Education. This report is our original work and the project has not formed the basis for the award of any degree/Diploma or any other similar titles.

By

SK. Karishma - 231FE04042

ACKNOWLEDGEMENT

This Minor Project-II program is a golden opportunity for learning and self-development. Consider ourselves very lucky and honored to have so many wonderful people lead us through in the completion.

We extend our sincere appreciation to the Project Guide **Mr. A. Siva Rao**, for granting us permission to undertake the Program and for his continuous support and cooperation throughout the duration of our Minor Project-l program.

It is a great pleasure for us to express our sincere thanks **Dr.S.K. Satpathy**, Director, Directorate of Polytechnic Education of VFSTR Deemed to be a university, for providing us with an opportunity to do our Minor Project-1 program.

We extend our wholehearted gratitude to all our faculty members, programmers, and technicians of the Directorate of Polytechnic Education who helped using our academics throughout the Minor Project-I program.

Finally, we wish to express thanks to our family members for the love and affection overseas and cheerful depositions, which are vital for sustaining the effort required for completing this work.

With Sincere Regards,

SK. Karishma - 231FE04042

ABSTRACT

The Leave Management System is a web-based application designed to automate and simplify the process of applying, approving, and managing student leave requests in an educational institution. Traditionally, leave applications are processed manually, leading to delays, errors, and lack of proper record maintenance. This system provides an efficient, user-friendly, and paperless solution that allows students to apply for leaves online, and counsellors to review and manage leave requests seamlessly. The project is developed using HTML, CSS, and JavaScript for the front-end interface, ensuring a responsive and visually appealing design. The back-end is implemented using PHP for server-side logic and MySQL for reliable data storage and management. The system enables students to submit leave applications within a specific time window and view their leave status, while counsellors can track, approve, or reject student leave requests based on the records stored in the database. By automating the leave approval workflow, the system enhances transparency, saves administrative time, and maintains an organized digital record of student leaves. This application can be easily deployed using XAMPP server, making it accessible on local or institutional networks.

TABLE OF CONTENTS

LIST OF FIGURES

CHAPTER 1: INTRODUCTION

1.1 Project Background

The process of managing student leave requests in educational institutions is often carried out manually, which can lead to inefficiencies, delays, and loss of records. Traditional methods involve filling out paper forms and physically submitting them to counsellors or administrators, making it difficult to maintain an organized and transparent record of leave applications.

To address these limitations, the Online Leave Management System has been developed to automate and streamline the leave application and approval process. This system enables students to apply for leave online and allows counsellors to review and update the leave status efficiently.

By incorporating both a user-friendly frontend interface and a robust backend database, the system ensures data consistency, accuracy, and easy access to leave history. The automated approach minimizes manual work, enhances communication between students and counsellors, and provides a faster, more reliable way to manage leave-related activities within the institution.

1.2 Objective

The main objective of this project is to create a simple and efficient platform for managing student leave requests digitally. It aims to allow students to apply for leave online, provide counsellors with an easy approval system, and introduce slot-based checking to avoid unnecessary delays. The project also ensures clear status updates, and proper digital records of all leave applications, making the process transparent and convenient for both students and faculty.

CHAPTER 2: SYSTEM REQUIREMENTS

Software Requirements: -

- Windows 10 Pro

Technology Used: -

- Front End: - HTML, Java Script, CSS
- Back End: - PHP
- Database: - MySQL Server: XAMPP / Localhost or Cloud Deployment

Hardware Requirements: -

- Processor: Intel i3
- RAM: 4 GB minimum
- Hard Disk: 250 GB or more
- Device: Desktop / Laptop with Internet access

CHAPTER 3: PROPOSED SYSTEM

Proposed system:

A web-based portal where students can apply for leave online and counsellors can approve or reject requests during fixed time slots. The system provides real-time status updates, reducing manual communication.

Advantages:

- ✓ Saves time and effort
- ✓ Transparent and reliable process
- ✓ Digital leave records maintained
- ✓ Easy access anytime, anywhere

CHAPTER 4: EXISTING SYSTEM

Existing system:

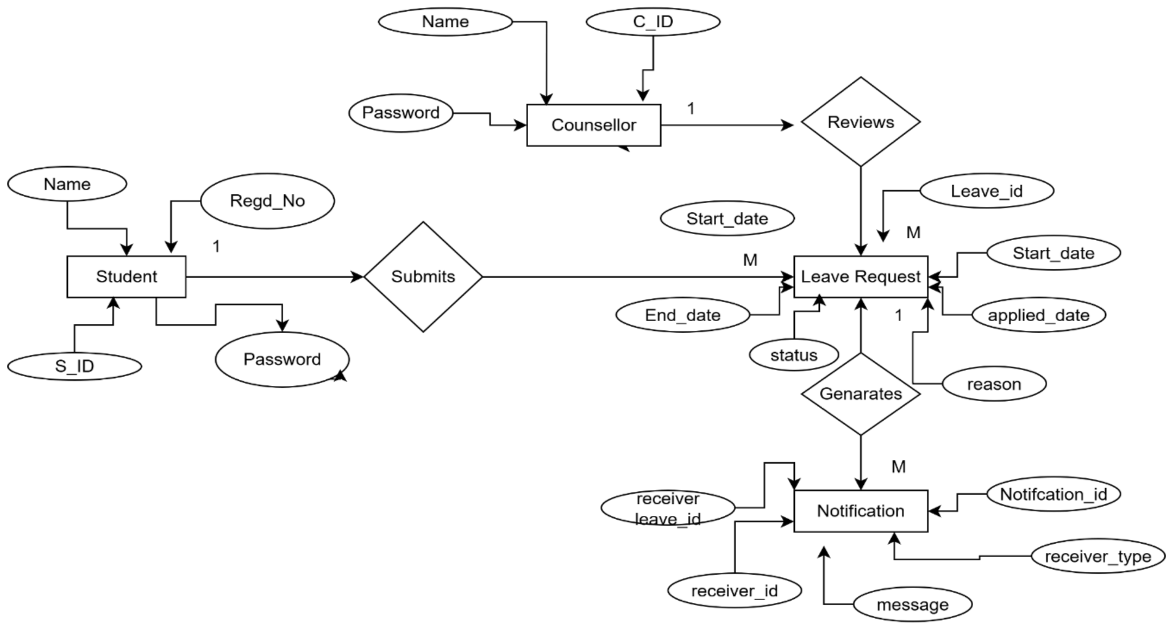
In the current system, students usually inform their counsellors about leave through phone calls, messages, or in person. Counsellors manually note down the details and give approval. This method is time-consuming and lacks proper record maintenance.

Disadvantages:

- No centralized digital record of student leaves.
- Counsellors waste time responding to multiple calls/messages.
- Risk of miscommunication or missed information.
- Notifications or updates are not systematic.
- Difficult to track leave history for future reference.

CHAPTER 5: ANALYSIS &DESIGN

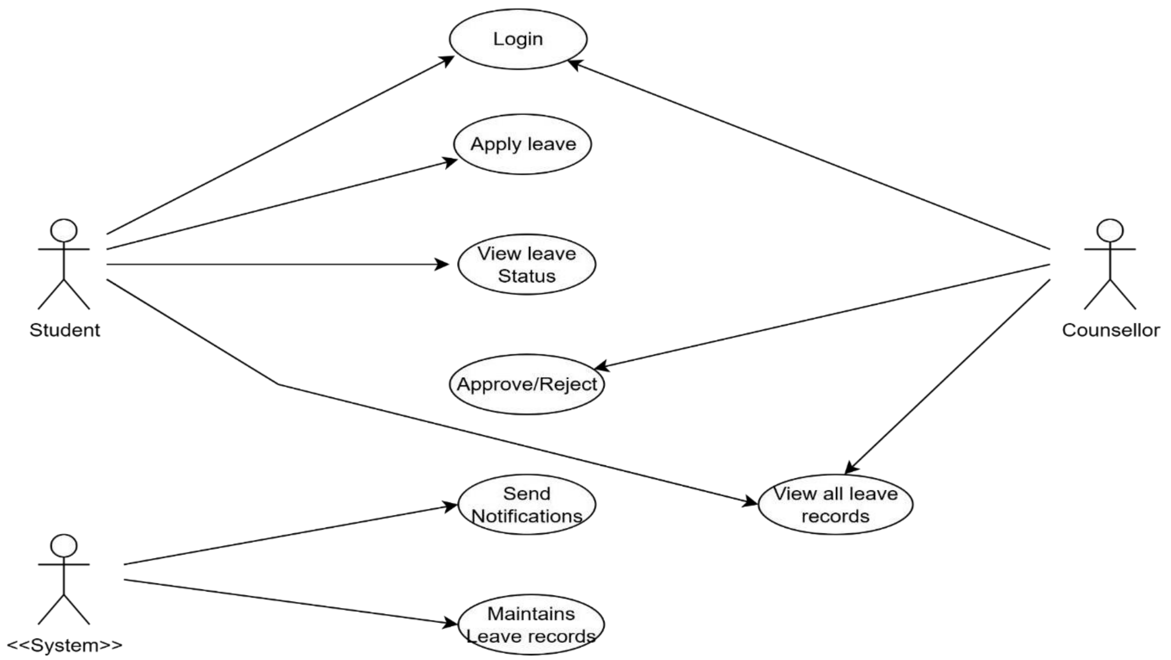
5.1 ER – DIAGRAM: -



5.2 USE CASES DIAGRAM

Use Cases Identified:

- Login
- Apply for Leave
- View Leave Status
- View Leave History
- Receive Notifications
- View Pending Leave Requests
- Approve or Reject Leave
- Send Status Notifications
- View All Leave Records
- Maintain Digital Records
- Generate Notifications

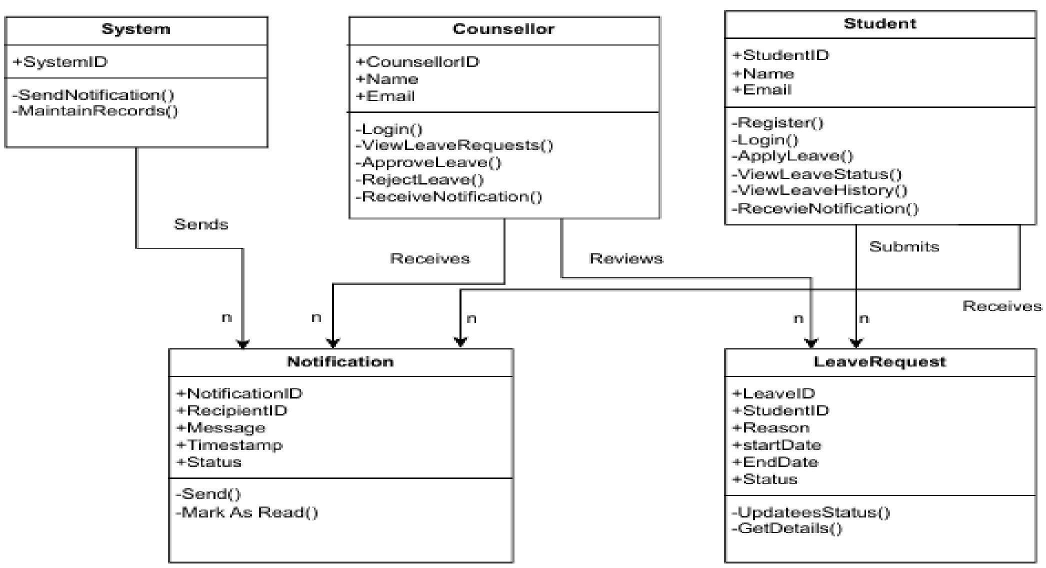


5.3 CLASS DIAGRAM

A class diagram is a type of UML (Unified Modelling Language) diagram used to represent the static structure of a system. It shows the system's classes, their attributes, methods (operations), and the relationships between objects. Class diagrams help developers understand and design the system architecture by visualizing how different parts of the application interact.

It shows the attributes, classes, functions, and relationships to give an overview of the software system. It constitutes class names, attributes, and functions in a separate compartment that helps in software development. Since it is a collection of classes, interfaces, associations, collaborations, and constraints, it is termed as a structural diagram.

Class Diagram of Parking management System:



CHAPTER 6: IMPLEMENTATION

6.1 Coding

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Leave Management System</title>
<style>
   body {
       font-family: "Poppins", sans-serif; margin: 0; padding: 0; height: 100vh;
       background: linear-gradient (135deg, #87CEEB, #00BFFF); display: flex;
       justify-content: center; align-items: center;
   }
  .container {
       text-align: center; background: white; padding: 50px 70px;
       border-radius: 20px; box-shadow: 0 4px 20px rgba(0,0,0,0.2);
   }
  h1 {
       color: #007ACC; font-size: 30px; margin-bottom: 40px;
   }
 button {
       width: 200px; padding: 12px; margin: 15px; border: none;
       border-radius:25px; font-size: 18px; font-weight: 600; cursor: pointer;
       color: white; background: linear-gradient (135deg, #00BFFF, #1E90FF);
       transition: transform 0.2s ease, background 0.3s ease;
   }
```

```
   button:hover {
            transform: scale(1.05);background: linear-gradient(135deg, #1E90FF,
#00BFFF);
   }
</style></head>
<body>
<div class=”container”>
   <h1>Leave Management System</h1>
 <button
onclick=”window.location.href=’register.html’”>Register</button><br>
   <button onclick=”window.location.href=’login.html’”>Login</button>
</div>
</body>
</html>
```

register.html

```
<!DOCTYPE html>
<html lang=”en”>
<head>
<meta charset=”UTF-8”>
<meta name=”viewport” content=”width=device-width, initial-scale=1.0”>
<title>Register | Leave Management System</title>
<style>
   body {
       font-family: “Poppins”, sans-serif; margin: 0; padding: 0;
       background: linear-gradient (to right, #87CEFA, #B0E0E6);display: flex;
       justify-content: center; align-items: center; height: 100vh;
   }
```

```
  .container {
      background: white; padding: 40px 50px; border-radius: 15px;
      box-shadow: 0 0 15px rgba(0,0,0,0.2);width: 360px; text-align: center;
  }
 h1 {
      color: #1E90FF; margin-bottom: 25px;font-size: 28px;
  }

  input, select {
      width: 100%; padding: 10px; margin: 10px 0; border: 1px solid #ccc;
      border-radius: 8px;font-size: 15px;
  }
button {
      width: 100%; padding: 10px; background-color: #1E90FF; color: white;
      font-size: 16px;font-weight: 600; border: none;border-radius: 8px;
      cursor: pointer; margin-top: 10px;
  }
button: hover {
      background-color: #4682B4;
  }
p {
      margin-top: 15px;color: #333;font-size: 14px;
  }
a {
      color: #1E90FF; font-weight: 600; text-decoration: none;
  }
```

```
    a: hover {
        text-decoration: underline;
    }
</style>
</head>
<body>
<div class="container">
    <h1>Leave Management System</h1>
    <form onsubmit="handleRegister(event)">
        <select id="role" required>
            <option value="">Select Role</option>
            <option value="student">Student</option>
            <option value="counsellor">Counsellor</option>
        </select>
 <input type="text" id="userid" placeholder="Enter Register/ID Number" required><input type="password" id="password" placeholder="Enter Password" required> <button type="submit">Register</button></form>
<p>Already have an account? <a href="login.html">Login now</a></p>
</div>
<script>
function handleRegister(event) {
    event.preventDefault();
    alert("Registration successful! Please login to continue.");
    window.location.href = "login.html";
}
</script>
</body>
</html>
```

login.html:

```
<!DOCTYPE html>
<html lang=”en”>
<head>
<meta charset=”utf-8” />
<meta name=”viewport” content=”width=device-width,initial-scale=1” />
<title>Leave Management System – Login</title>
<style>
  :root{--sky1:#d7ecff;--sky2:#bfe7ff;--accent:#0d6efd}
 body{margin:0;font-family:Arial ,Helvetica,sans-serif;background:linear-gradient(180deg,var(--sky1),var(--sky2));min-height:100vh; display: flex; align-items:center;justify-content: center}
 .card{width:360px; background: #fff; padding:22px; border-radius:12px; box-shadow:0 8px 30px rgba(17,40,80,0.08); text-align: center}
  h1{margin:0 0 14px;color:var(--accent); font-size:20px}
 label {display:block;text-align: left; margin:10px6px; color:#333; font-weight:600;font-size:13px}
 select,input{width:100%;padding:10px;border-radius:8px;border:1px solid #ddd;font-size:14px}
button{margin-top:14px;width:100%; padding:11px; border-radius:8px;border:none; background:linear-gradient(90 deg, var(--accent),#00a3ff);color:#fff;font-weight:700;cursor:pointer}
  .small{margin-top:12px;font-size:13px;color:#555}
  a{color:var(--accent);text-decoration:none;font-weight:600}
</style>
</head>
<body>
  <div class=”card”>
    <h1>Leave Management System</h1>
```

```
   <label for=”role”>Role</label>
<select
id=”role”><option>Student</option><option>Counsellor</option></select>
  <label for=”id”>Register / Counsellor ID</label>
    <input id=”id” placeholder=”Enter ID”>  <label for=”pass”>Password</label>
    <input id=”pass” type=”password” placeholder=”Enter password”>
 <button onclick=”login()”>Login</button>
<div class=”small”>Don’t have an account?
 <a href=”register.html”>Register</a></div>
  </div>
<script>
function login(){
  const role = document.getElementById(‘role’).value;
  // Demo routing based on role – replace with backend auth later
  if(role===’Counsellor’) window.location.href=’counsellor_dashboard.html’;
  else window.location.href=’student_dashboard.html’;
}
</script>
</body>
</html>
```

student_dashboard.html:

```
<!DOCTYPE html>
<html lang=”en”>
<head>
<meta charset=”utf-8” />
<meta name=”viewport” content=”width=device-width,initial-scale=1” />
<title>Dashboard</title>
<style>
```

```
:root{--sky1:#e7f6ff;--sky2:#d7efff;--accent:#1e90ff}

body{margin:0;font-family: Arial, Helvetica, sans-serif; background: linear-gradient(180deg,var(--sky1),var(--sky2)); min-height:100vh}

header {display: flex; align-items: center; justify-content: space-between; padding:16px 28px}

.title{font-weight:700; color :#074f9c} /* top-left “Dashboard” */

.logout{background: #ff5c5c; border:none; color:#fff; padding:8px 14px; border-radius:8px; cursor:pointer}

.center{display:flex;flex-direction: column; align-items: center; justify-content: center; height: calc (100vh – 72px); text-align: center}

.welcome{font-size:20px; color: #074f9c; margin-bottom:18px; font-weight:700}

.buttons{display:flex;gap:18px}

.btn{background: linear-gradient (90deg, var(--accent),#00aaff); color :#fff; padding:14px 28px; border-radius:10px;border:none; font-weight:700; cursor: pointer}

.panel{width:80%;max-width:820px; margin:18px auto;background:#fff;padding:22px;border-radius:12px; box-shadow:0 10px 30px rgba(7,79,156,0.06);display:none}

h2{color:var(--accent);margin-top:0}

label{display:block;margin:10px 0 6px;color:#333;font-weight:600}

input,textarea{width:100%;padding:10px;border-radius:8px;border:1px solid #ddd;font-size:14px}

textarea{min-height:120px;resize:vertical}

.submit{margin-top:12px; background: linear-gradient (90deg, var (--accent),#00aaff);color :#fff; padding:10px 18px; border:none; border-radius:8px; cursor:pointer; font-weight:700}

table{width:100%;border-collapse:collapse;margin-top:12px}

th,td{padding:10px;border:1px solid #eee;text-align:left}

th{background:var(--accent);color:#fff}
```

```
  .back{margin-top:12px;     background:#eee; border: none;padding:8px 12px;border-radius:8px;cursor:pointer}
  .status{margin-top:12px;padding:10px;border-radius:8px;background:#f0f8ff;color:#005bb5;font-weight:600}
</style>
</head>
<body>
<header>
  <div class=”title”>Dashboard</div>
  <button class=”logout” onclick=”logout()”>Logout</button>
</header>
<div class=”center”>
  <div class=”welcome”>Welcome 231FE04027</div>
  <div class=”buttons”>
    <button class=”btn” onclick=”openApply()”>Apply Leave</button>
    <button class=”btn” onclick=”openPast()”>View My Past Leaves</button>
  </div>
</div><script>
function openApply(){ window.location.href=’apply_leave.html’; }
function openPast(){ window.location.href=’view_leaves.html’; }
function logout(){ alert(‘Logged out’); window.location.href=’login.html’; }
</script></body>
</html>
```

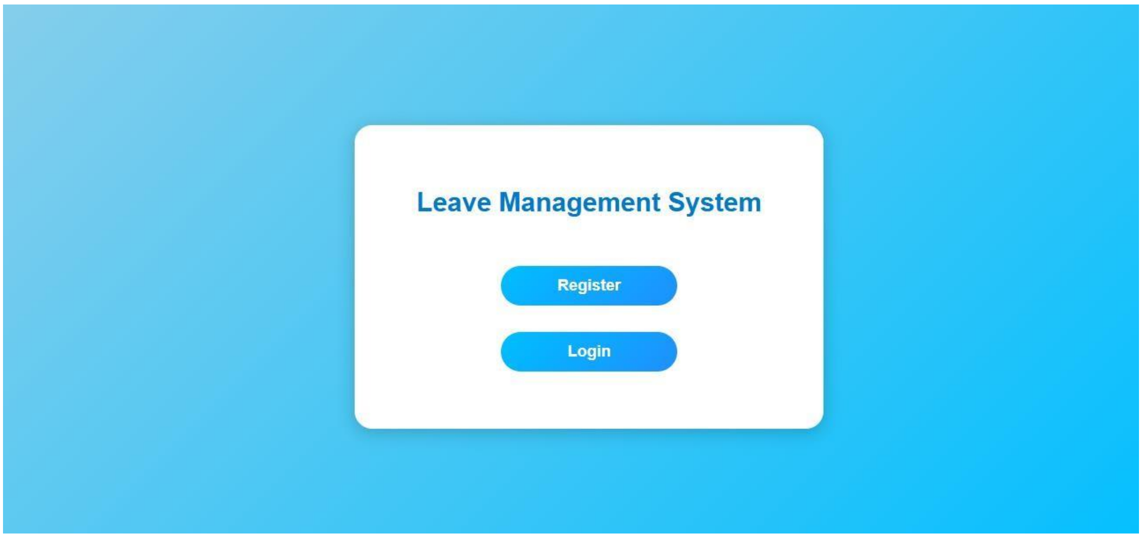
counsellor_dashboard:

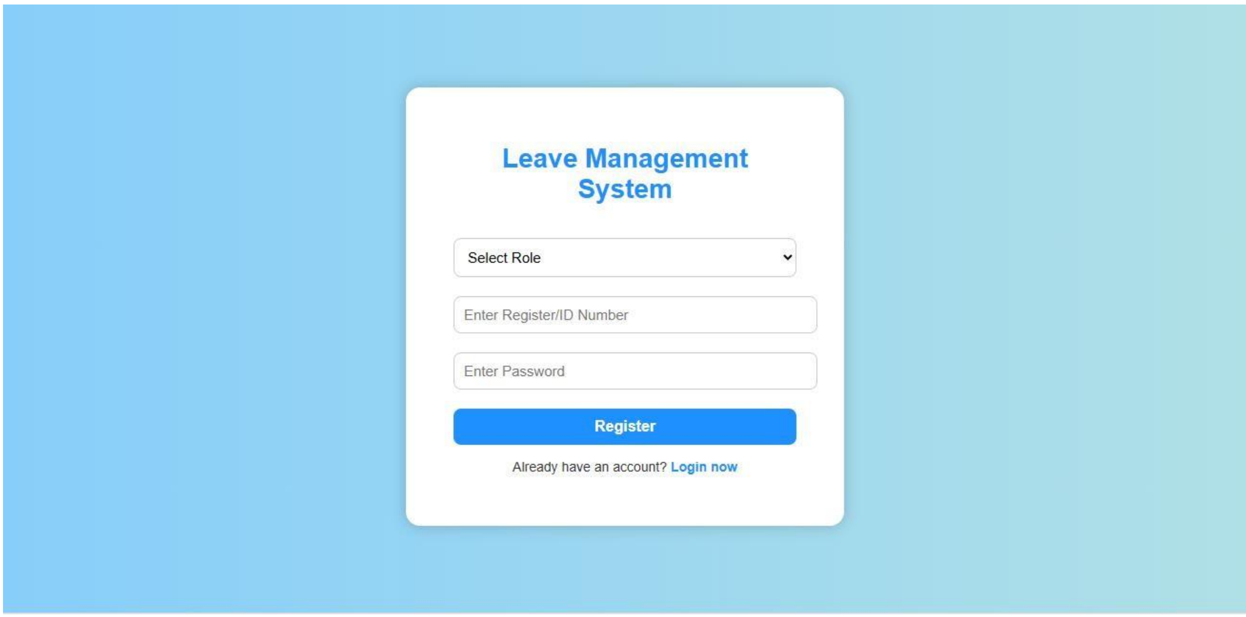
```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8" />
```

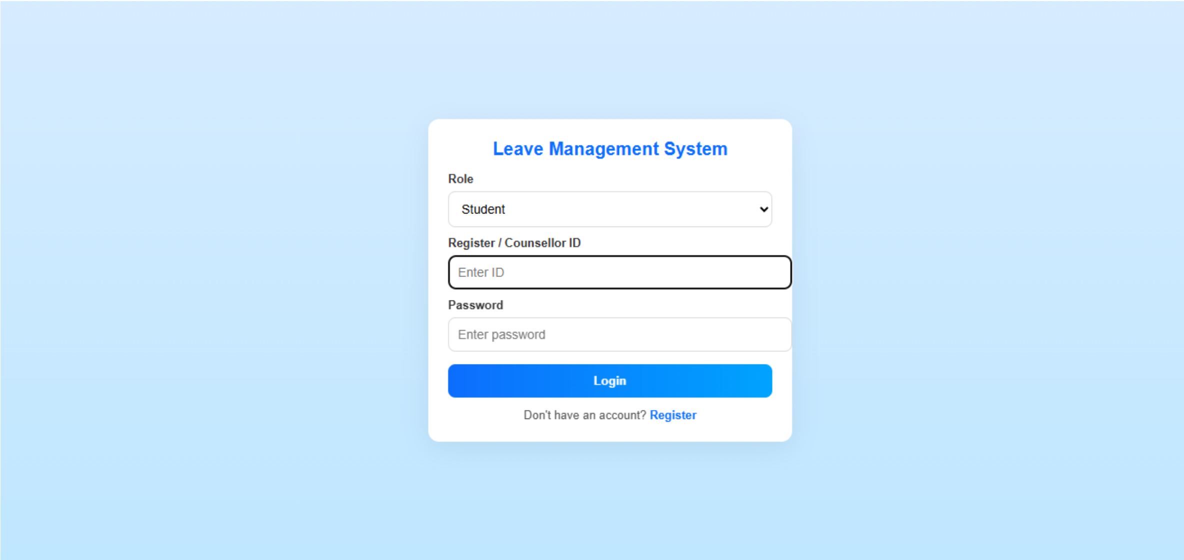
```
<meta name="viewport" content="width=device-width, initial-scale=1" />
<title>Dashboard</title>
<style>
  :root{--sky1:#e7f8ff;--sky2:#d7f0ff;--accent:#00a3ff}
  body{margin:0;font-family: Arial, Helvetica, sans-serif; background: linear-gradient(180deg,var(--sky1),var(--sky2));min-height:100vh}
  header{display:flex;align-items:center;justify-content:space-between;padding:16px 24px}
  .title{font-weight:700; color:#075a9c}
  .logout {background: #ff5c5c; border: none; color: #fff; padding:8px 14px; border-radius:8px; cursor: pointer}
  .center{display: flex; flex-direction: column; align-items: center; justify-content: center; height: calc (100vh - 72px)}
  .welcome{font-size:20px; color: #075a9c; margin-bottom:18px; font-weight:700}
  .buttons{display:flex;gap:18px}
  .btn{background: linear-gradient (90deg, var (--accent), #00c0ff); color:#fff; padding:14px 28px; border-radius:10px;border:none;font-weight:700;cursor:pointer}
  .panel{width:80%; max-width:900px; margin:18px auto;background:#fff; padding:22px;border-radius:12px; box-shadow:0 10px 30px rgba(7,90,156,0.06);display:none}
  table{width:100%;border-collapse:collapse;margin-top:12px}
  th,td{padding:10px;border:1px solid #eee;text-align:left}
  th{background:linear-gradient(90deg,var(--accent),#00c0ff);color:#fff}
  .action{display:flex;gap:8px}
  .approve{background:#28a745;color:#fff;padding:6px 10px;border-radius:8px;border:none;cursor:pointer}
  .reject{background:#e53935;color:#fff;padding:6px 10px;border-radius:8px;border:none;cursor:pointer}
```

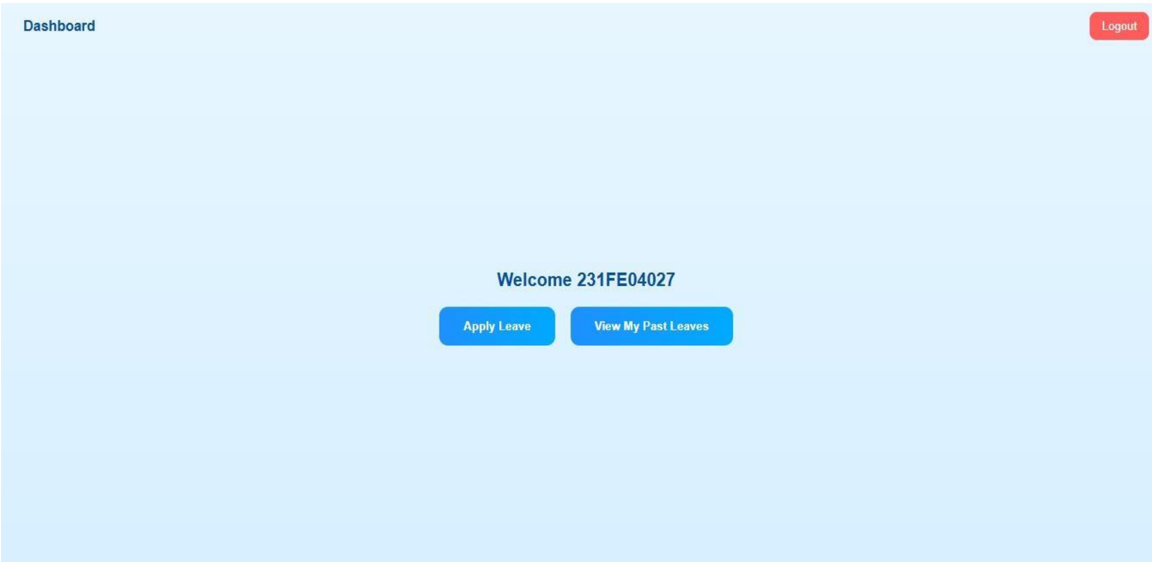
```
  .back{margin-top:12px;background:#eee;border:none;padding:8px 12px;border-radius:8px;cursor:pointer}
  .no-data{margin-top:12px;color:#666;font-weight:600}
  .row-approved{background:rgba(40,167,69,0.12)}
  .row-rejected{background:rgba(229,57,53,0.12)}
</style>
</head>
<body>
<header>
  <div class="title">Dashboard</div>
  <button class="logout" onclick="logout()">Logout</button>
</header>
<div class="center">
  <div class="welcome">Welcome Counsellor</div>
  <div class="buttons">
        <button class="btn" onclick="gotoPending()">View Pending Requests</button>
    <button class="btn" onclick="gotoPast()">View Past Leaves</button>
  </div>
</div>
<script>
function logout(){ alert('Logged out'); window.location.href='login.html'; }
function gotoPending(){ window.location.href='view_pending.html'; }
function gotoPast(){ window.location.href='counsellor_view_past.html'; }
</script>
</body></html>
```

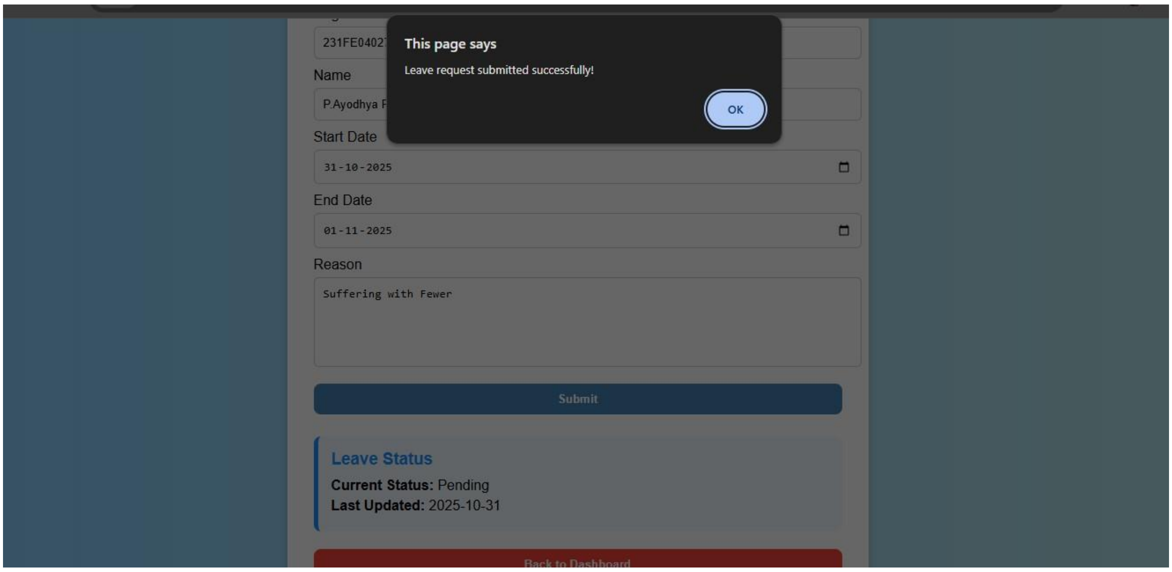
6.2 OUPUTS

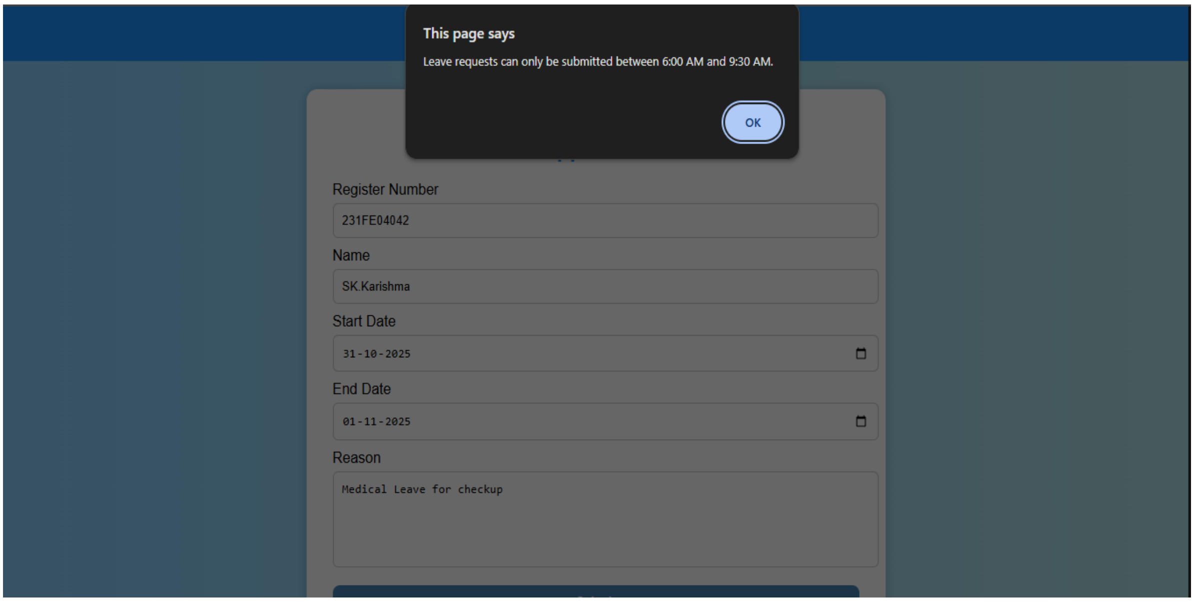

Leave Management System
Register
Login

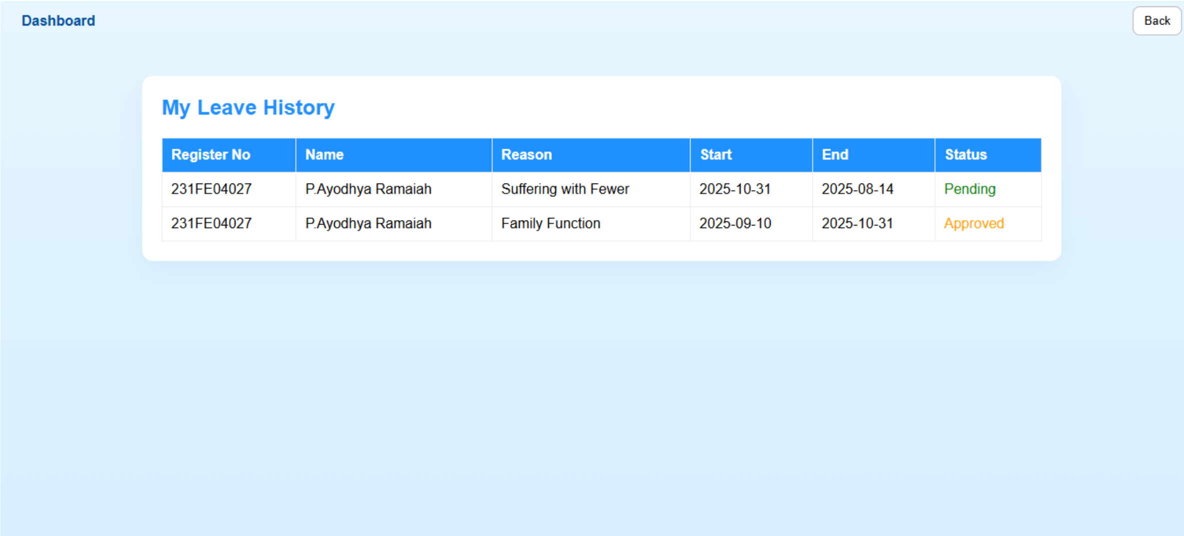

Leave Management System
Select Role
Enter Register/ID Number
Enter Password
Register
Already have an account? Login now

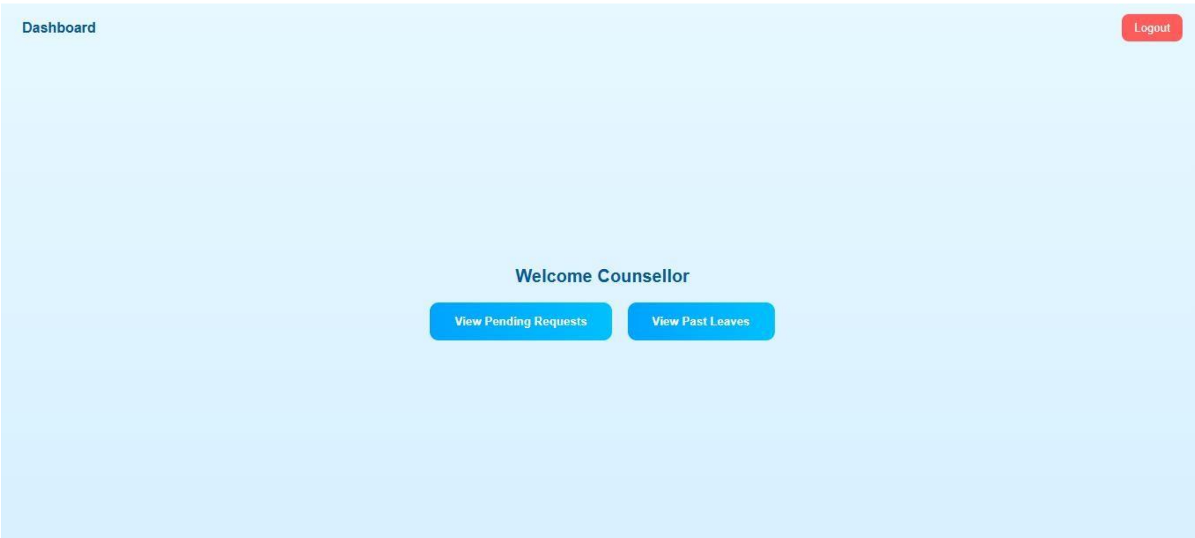

Leave Management System
Role
Student
Register / Counsellor ID
Enter ID
Password
Enter password
Login
Don't have an account? Register

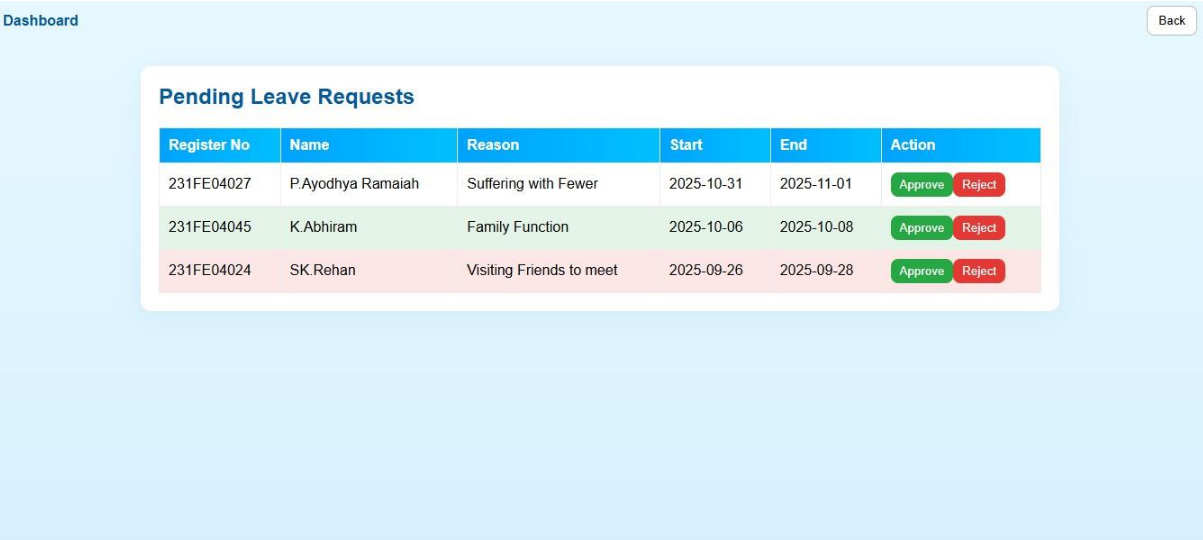

Dashboard
Logout
Welcome 231FE04027
Apply Leave
View My Past Leaves


This page says
Leave request submitted successfully!
OK
231FE0402
Name
P.Ayodhya
Start Date
31-10-2025
End Date
01-11-2025
Reason
Suffering with Fewer
Submit
Leave Status
Current Status: Pending
Last Updated: 2025-10-31
Back to Dashboard

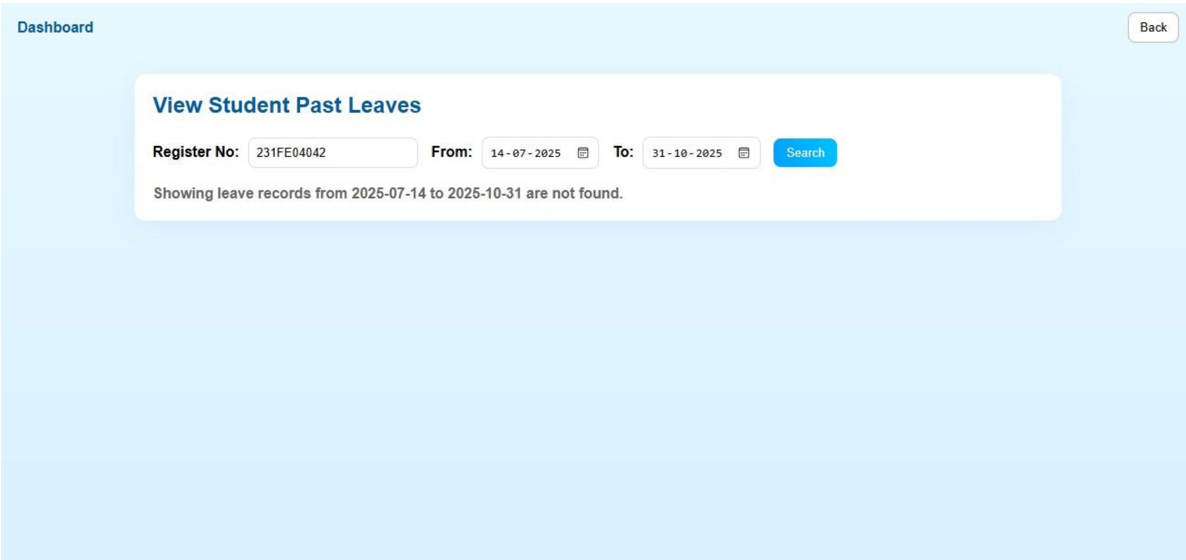

This page says
Leave requests can only be submitted between 6:00 AM and 9:30 AM.
OK
Register Number
231FE04042
Name
SK.Karishma
Start Date
31-10-2025
End Date
01-11-2025
Reason
Medical Leave for checkup


Dashboard
Back
My Leave History
Register No | Name | Reason | Start | End | Status
231FE04027 | P.Ayodhya Ramaiah | Suffering with Fewer | 2025-10-31 | 2025-08-14 | Pending
231FE04027 | P.Ayodhya Ramaiah | Family Function | 2025-09-10 | 2025-10-31 | Approved


Dashboard
Logout
Welcome Counsellor
View Pending Requests
View Past Leaves



Register No	Name	Reason	Start	End	Action
231FE04027	P.Ayodhya Ramaiah	Suffering with Fewer	2025-10-31	2025-11-01	Approve Reject
231FE04045	K.Abhiram	Family Function	2025-10-06	2025-10-08	Approve Reject
231FE04024	SK.Rehan	Visiting Friends to meet	2025-09-26	2025-09-28	Approve Reject



CHAPTER 7: TESTING

TESTING:

System testing of software or hardware is testing conducted on a complete, integrated system to evaluate the system's compliance with its specified requirements. System testing falls within the scope of black box testing, and as such, should require no knowledge of the inner design of the code or logic. As a rule, system testing takes, as its input, all the "integrated" software components that have passed integration testing and also the software system itself integrated with any applicable hardware system(s). The purpose of integration testing is to detect any inconsistencies between the software units that arc integrated together (called assemblages) or between any of the assemblages and the hardware. System testing is a more limited type of testing; it seeks to detect defects both within the "inter-assemblages" and also within the system.

7.1 UNIT TESTING

In computer programming, unit testing is a software testing method by which individual units of source code, sets of one or more computer program modules together with associated control data, usage procedures, and operating procedures, are tested to determine whether they are fit for use. Intuitively, one can view a unit as the smallest testable part of an application. In procedural programming, a unit could be an entire module, but it is more commonly an individual function or procedure. In object-oriented programming, a unit is often an entire interface, such as a class, but could be an individual method. Unit tests are short code fragments created by programmers or occasionally by white box testers during the development process. It forms the basis for component testing.

Ideally, each test case is independent from the others. Substitutes such as method stubs, mock objects, fakes, and test harnesses can be used to assist testing a module in isolation. Unit tests are typically written and run by software developers to ensure that code meets its design and behaves as intended.

The goal of unit testing is to isolate each part of the program and show that the individual parts are correct. A unit test provides a strict, written contract that the piece of code must satisfy.

7.2INTEGRATION TESTING

The Integration testing (sometimes called integration and testing, abbreviated I&T) is the phase in software testing in which individual software modules are combined and tested as a group. It occurs after unit testing and before validation testing. Integration testing takes as its input modules that have been unit tested, groups them in larger aggregates, applies tests defined in an integration test plan to those aggregates, and delivers as its output the integrated system ready for system testing.

The purpose of integration testing is to verify functional, performance, and reliability requirements placed on major design items. These "design items", i.e. assemblages (or groups of units), are exercised through their interfaces using black box testing, success and error cases being simulated via appropriate parameter and data inputs. Simulated usage of shared data areas and inter-process communication is tested and individual subsystems are exercised through their input interface. Test cases are constructed to test whether all the components within assemblages interact correctly, f or example across procedure calls or process activations, and this is done after testing individual modules, i.e. unit testing. overall idea is a "building block" approach, in which verified assemblages are added to a verified base which is then used to support the integration testing of further assemblages.

7.3 VALIDATION TESTING

- Determining if the system complies with the requirements and performs functions for which it is intended and meets the organization’s goals and user needs.
- Validation is done at the end of the development process and takes place after verifications are completed.
- It answers the question like: Am I building the right product?
- Am I accessing the right data (in terms of the data required to satisfy the requirement).
- It is a High level activity.
- Performed after a work product is produced against established criteria ensuring that the product integrates correctly into the environment.
- Determination of correctness of the final software product by a development project with respect to the user needs and requirements.

A product can pass while verification, as it is done on the paper and no running or functional application is required. But, when same points which were verified on the paper is actually developed then the running application or product can fail while validation. This may happen because when a product or application is built as per the specification but these specifications are not up to the mark hence they fail to address the user requirements.

Advantages of Validation

- During verification if some defects are missed then during validation process it can be caught as failures.
- If during verification some specification is misunderstood and development had happened, then during validation process while executing that functionality the difference between the actual result and expected result can be understood.
- Validation is done during testing like feature testing, integration testing, system testing, load testing, compatibility testing, stress testing, etc.

Validation is basically done by the testers during the testing. While validating the product

if some deviation is found in the actual result from the expected result then a bug is reported or an incident is raised. Not all incidents are bugs. But all bugs are incidents.

Incidents can also be of type 'Question' where the functionality is not clear to the tester.

Hence, validation helps in unfolding the exact functionality of the features and helps the

testers to understand the product in much better way. It helps in making the product more

user friendly.

CHAPTER 8: CONCLUSION

Conclusion:

The Leave Management System simplifies and automates the traditional leave process in educational institutions. By providing separate dashboards for students and counsellors, the system enhances transparency, reduces paperwork, and saves time. Students can easily apply for leave, while counsellors can view and manage requests efficiently. Though the current version operates locally with limited features, it lays a strong foundation for future improvements such as real-time notifications, enhanced security, and cloud integration. With further development, the system can evolve into a complete digital leave management platform for educational and organizational use.

CHAPTER 9: FUTURE SCOPE

Future scope:

- Integration of email or SMS notifications to alert students about their leave status.
- Adding an admin module to manage counsellors, students, and overall leave data.
- Implementing AI-based analytics to predict student attendance trends and leave patterns.
- Providing mobile application support for better accessibility.
- Enabling cloud storage for secure and remote access to data.
- Introducing role-based authentication and multi-user login using secure credentials.
- Generating automated reports of student leave records for management review.

CHAPTER 10: REFERENCES

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